Customer Shopping Behavior Analysis

End-to-End Data Analytics Project | Python • MySQL • Power BI

Project overview

This project explores customer shopping behavior across 3,900 purchase records to examine purchase amounts, product categories, subscription status, shipping preferences, review ratings, and customer age groups. The workflow uses Python for data preparation, SQL for business questions, and Power BI for visual reporting.

Project workflow

Stage	Purpose
Python / pandas	Inspect, clean, standardize and enrich the dataset; load the prepared data to a relational database.
MySQL	Answer ten business questions using aggregations, subqueries, CASE logic, CTEs and window functions.
Power BI	Present customer KPIs and compare subscription, category and age-group patterns with interactive filters.

1. Python: data preparation and exploratory analysis

The supplied project report documents the following notebook steps: load data with pandas; inspect structure using `info()` and summary statistics using `describe()`; handle 37 missing review ratings by imputing the median rating within each product category; standardize column names to snake_case; create `age_group` and `purchase_frequency_days`; check redundancy between `discount_applied` and `promo_code_used`; and load the cleaned DataFrame into a SQL table for further analysis.

Representative Python workflow

```
# Core analysis pattern (adapted; update the CSV path for your environment)
import pandas as pd

df = pd.read_csv("customer_shopping_behavior.csv")
print(df.info())
print(df.describe(include="all"))

# Impute review ratings within each product category
df["review_rating"] = df.groupby("category")["review_rating"].transform(
    lambda s: s.fillna(s.median())
)

# Continue with the notebook's column standardization and feature engineering.
# Configure database credentials locally; never commit real passwords.
```

Note: this excerpt is an illustrative summary of the documented workflow, not a replacement for running the complete notebook. The original notebook and dataset should be executed locally to verify all cells and results.

2. SQL: business questions and query logic

The uploaded SQL script contains ten analyses. The table below summarizes the questions covered by the script.

#	Business question	SQL approach
Q1	Total purchase revenue by gender	GROUP BY gender; SUM(purchase_amount)
Q2	Discount users who spent above the overall average	WHERE discount_applied = 'Yes' and scalar AVG subquery
Q3	Five products with the highest average review rating	AVG(review_rating), ORDER BY descending, LIMIT 5
Q4	Average purchase amount for Standard vs Express shipping	Filter shipping types; GROUP BY shipping_type
Q5	Subscriber vs non-subscriber customer count, average spend and revenue	COUNT, AVG, SUM grouped by subscription_status
Q6	Five products with highest discount-use percentage	Conditional SUM / COUNT
Q7	Rule-based New, Returning and Loyal segments	CTE with CASE and previous_purchases thresholds
Q8	Top three purchased items within each category	CTE with ROW_NUMBER() window function

#	Business question	SQL approach
Q9	Subscription status among customers with >5 previous purchases	Filter repeat buyers and GROUP BY subscription_status
Q10	Revenue by age group	SUM(purchase_amount) grouped by age_group

SQL examples

```
-- Q1: Total purchase revenue by gender
SELECT gender, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY gender;

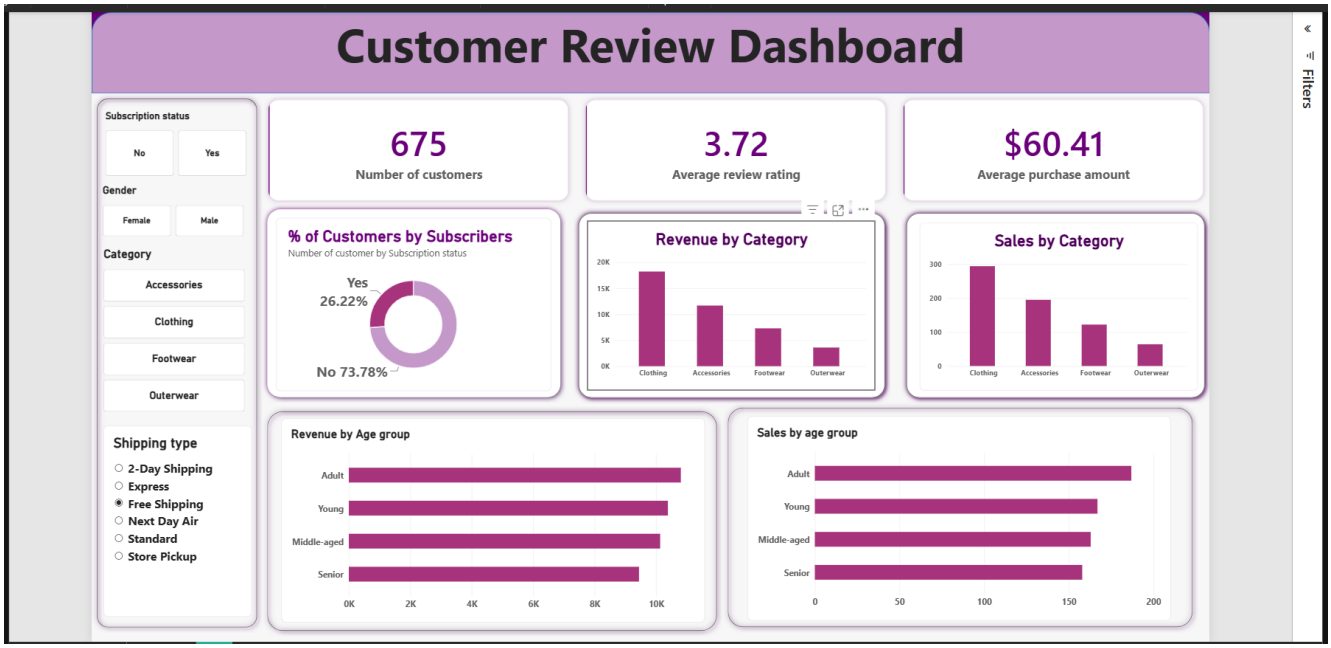
-- Q5: Compare subscription groups
SELECT subscription_status,
       COUNT(customer_id) AS customer_count,
       ROUND(AVG(purchase_amount), 2) AS average_spend,
       SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY subscription_status;

-- Q8: Rank the most-purchased items within each category
WITH item_counts AS (
  SELECT category, item_purchased, COUNT(*) AS total_orders,
         ROW_NUMBER() OVER (
           PARTITION BY category ORDER BY COUNT(*) DESC, item_purchased
         ) AS item_rank
  FROM customer
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
```

The complete SQL script should remain a separate file in the GitHub repository. Run it against the cleaned customer table and capture result grids for evidence. Query results reported in the source report should be rechecked against your own execution.

3. Power BI: dashboard and visual insights

The supplied dashboard screenshot shows a customer review dashboard with slicers for subscription status, gender, category, and shipping type. Visible KPIs in the captured state are 675 customers, a 3.72 average review rating, and a $60.41 average purchase amount. The subscription chart shows 26.22% Yes and 73.78% No. Clothing has the largest bar in the displayed revenue and sales-by-category visuals. Age-group charts compare Adult, Young, Middle-aged, and Senior groups.



Screenshot supplied by the author. The Free Shipping slicer is selected in this image; values describe the visible filtered state and may change with other selections.

Business recommendations

- Evaluate subscription benefits and test their effect on sign-ups and retention.
- Consider loyalty initiatives for repeat purchasers.
- Assess discount performance alongside margin or profit data before changing promotions.
- Feature highly rated and frequently purchased products, while checking sample sizes.
- Use age-group and shipping patterns to form testable marketing hypotheses rather than assuming causation.

Limitations, attribution and validation

This is descriptive analysis of observational data and does not establish causality. Purchase revenue should not be interpreted as profit. Customer segments are rule-based and depend on selected purchase-history thresholds. The project was developed with reference to a YouTube tutorial by Amlan Mohanty; retain attribution to the creator and verify the dataset's source and license before redistribution. The CSV dataset was not included with the uploaded files, so notebook execution and final query outputs still require local verification.